SAPLoc: An Advanced Multi-label Protein Subcellular Localization Prediction Model via Sparse Self-Attention Visual Encoding on the Immunohistochemistry Images

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Abstract:

Protein subcellular localization (PSL) prediction from high-resolution immuno-histochemistry (IHC) images is critical for deciphering protein functions and unraveling disease mechanisms. However, existing global attention-based models grapple with two pivotal challenges: heavy computational burdens arising from dense global attention and insufficient capture of multi-scale subcellular features, which is particularly pronounced when processing large IHC images. To tackle these issues, we propose SAPLoc, a novel multi-label PSL prediction model that embeds sparse self-attention (SSA) into the visual encoding of IHC images. Its core innovations lie in two aspects: first, by substituting dense global attention with SSA, SAPLoc adaptively hones in on informative regions while mitigating redundant computations, thereby reducing the complexity of global attention-based models and enabling efficient processing of high-resolution IHC images without excessive downsampling that would erode critical details. Second, SAPLoc leverages a hierarchical architecture with adjustable sparse ratios across stages, which ensures robust feature extraction for subcellular structures of varying sizes, including both small organelles and large cellular compartments, by balancing the preservation of local details and integration of global context. Experimental results on the Vislocas dataset demonstrate that SAPLoc achieves competitive performance across multiple evaluation metrics for PSL. By bridg- ing the gap between the demand for high-resolution details and computational feasibility, SAPLoc provides a scalable solution for precise PSL analysis in large biological images.

1 Introduction

Protein subcellular localization (PSL), defined as the process by which the residence of proteins within cells is determined, lies at the core of understanding cellular function, as the spatial distribution of a protein directly dictates its biological role [1]. It is

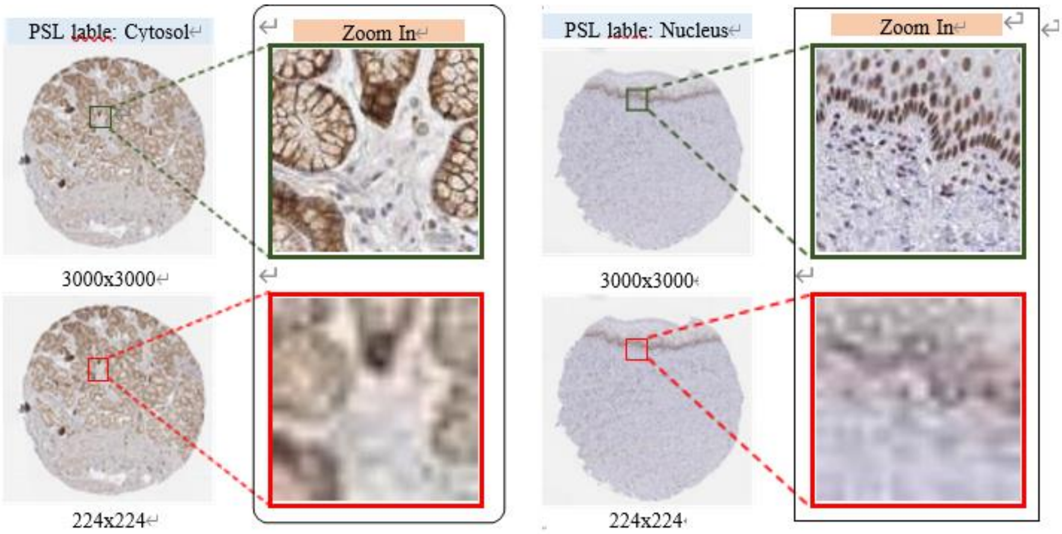


Fig. 1 Visual comparison between the original high-resolution IHC image (3000 × 3000 pixels) and the downsampled image (224 × 224 pixels).

well established that abnormal localization disrupts signaling pathways and molecular interactions, which underpin diseases ranging from neurodegenerative disorders such as Alzheimer's [2] to cancers [3]. Thus, accurate PSL prediction is pivotal for unraveling disease mechanisms and for the development of targeted therapies [4].

Traditional determination of PSL is reliant on wet-lab experiments, which are labor-intensive, costly, and difficult to scale with the exponential growth of protein data that is generated by high-throughput sequencing and imaging technologies [5, 6]. This shift toward computational methods—categorized into two primary types, those based on amino acid sequences and those based on images—is driven by such limita- tions. Of these two categories, sequence-based methods (e.g., Deeploc [7], Prot-SCL [8]) leverage protein primary structures to infer localization. While successful, they fail to capture dynamic protein translocations—critical for identifying cancer biomark- ers—and overlook spatial context within cellular environments [9]. In contrast, image- based methods directly analyze high-resolution microscopic images, offering intuitive visualization of protein distribution and dynamic localization changes. Among these, immunohistochemistry (IHC) images—abundantly available in databases like the Human Protein Atlas (HPA) [10]—are particularly valuable: their tissue-level res- olution preserves spatial context between cells, and their clinical relevance makes findings directly translatable to disease research. Early IHC-based PSL methods rely on handcrafted features, such as Haralick texture metrics [11] or local binary patterns (LBP) [12]. These approaches, however, are limited by subjective feature design and poor generalization to complex cellular environments [13]. The rise of deep learning (DL) has transformed this field: CNN-based feature extraction for PSL is pioneered by Shao et al. [14]; self-attention is integrated by Long et al. [15] to aggregate CNN features; and recent works like MSTLoc adopt Vision Transformers (ViTs)—a type of model that processes images as sequences of patches—to capture multi-scale patterns.

Despite these advances, DL-based models for IHC-based PSL face two critical lim- itations that hinder their performance and practicality. First, computational barriers arise with high-resolution imagery: IHC images often exceed 3000 × 3000 pixels, a size far larger than that of natural images. To reduce computational load, existing meth- ods downsample these images, but this process irreversibly loses fine-grained details, such as subtle protein aggregates in mitochondria or membrane-bound signaling com- plexes, which directly determine subcellular localization labels—that are critical for accurate localization (comparisons of original vs. downsampled IHC images are shown in Fig. 1). Second, there is an inadequate capture of local-global feature tradeoffs: current DL-based models, particularly Vision Transformers with dense global atten- tion, prioritize holistic image patterns but struggle to focus on localized subcellular structures. This imbalance impairs discrimination between morphologically similar organelles: for example, dense global attention may dilute signals from the endoplas- mic reticulum (a network of membranous tubules) amid the cytoplasm (a larger, more diffuse compartment), leading to misclassification.

To address these limitations, we propose SAPLoc, a novel multi-label PSL predic- tion model tailored for high-resolution IHC images. The key innovations of this study are as follows:

1. **First application of sparse self-attention (SSA) in PSL modeling**: As the first PSL prediction model to integrate SSA, SAPLoc replaces dense global atten- tion with this mechanism to adaptively focus on biologically relevant regions (e.g., organelle-bound protein signals) while suppressing redundant background interactions. This innovation reduces computational complexity by up to 85% compared to standard dense attention models (e.g., ViT-B with global attention), enabling direct processing of full-resolution IHC images without downsampling.

2. **Hierarchical multi-scale encoding module with adjustable sparsity ratios**: SAPLoc employs a staged architecture where sparse ratios are tuned across layers, with lower layers preserving fine-grained local details that are critical for identifying small organelles like mitochondria, and higher layers integrating global context that is essential for characterizing large compartments like the nucleus. This hierarchical balance ensures robust feature extraction across diverse subcellular structures.

In summary, SAPLoc bridges the gap between high-resolution detail preservation and computational feasibility, offering a scalable solution for precise PSL analysis.

2 Results

2.1 Implementation Details

To fully exploit the features of high-resolution images while maintaining computational feasibility, the model was trained with an input resolution of 3000 ×3000 pixels. To mit- igate overfitting, the following data augmentation techniques were applied exclusively to the training set: random horizontal/vertical flipping and rotation within a range of $+15^{\circ}$ (each with a probability of 0.5). Additionally, all images were standardized using the channel-wise mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. The model was trained using the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with an initial learning rate of 1×10^{-4} and a weight decay coefficient of 5×10^{-2} to enhance regularization. The learning rate scheduling strategy involved two stages: linear warm-up over the first 10 epochs to gradually adjust the learning rate, followed by cosine annealing. The model was configured with a batch size of 1 and 80 training epochs. Hyperparameters across the four stages were set as follows: feature dimen- sions $[C_1, C_2, C_3, C_4]$ = [64, 128, 320, 512]; patch sizes $[P_1, P_2, P_3, P_4]$ = [5, 15, 30, 60]; number of layers $[L_1, L_2, L_3, L_4]$ = [5, 8, 20, 7]; sparsity rates for the third stage $[S_1, S_2, S_3, S_4]$ = [10, 5, 2, 1]; and sparsity rates for the fourth stage $[S_5, S_6]$ = [5, 1]. All experiments were conducted in a distributed training environment equipped with mul- tiple V100-SXM2-32G GPUs, utilizing a data parallelism strategy to improve training efficiency.

2.2 Evaluation Metrics

Following the evaluation protocols outlined in [3, 16], we utilize both label-based and example-based metrics for multi-label PSL classification. Label-level metrics include Label Accuracy (LA), Label Precision (LP), Label Recall (LR), and Label F1- score (LF1). For example-level metrics, we consider Subset Accuracy (SA), Example Accuracy (EA), Example Precision (EP), Example Recall (ER), and Example F1- score (EF1). Among these, Subset Accuracy (SA) is the most stringent and widely emphasized, as it requires all labels in a multi-label prediction to be correctly identified.

2.3 Baselines

To ensure a comprehensive evaluation, SAPLoc is compared against 13 existing models, which are categorized into four groups: 1) PSL-specific models (£): DeeP- SLoc [16], GraphLoc [17], and Vislocas [3], which are specifically designed for protein subcellular localization tasks and used as direct benchmarks to assess task-specific per- formance. 2) Popular CNNs (t): ResNet18 [18], ResNet50 [18], and DenseNet121 [19], which are representative CNNs widely adopted in computer vision and utilized to evaluate whether general visual feature extractors can be effectively adapted to high-resolution biomedical imaging scenarios. 3) Vision Transformers (‡): ViT [20], SwinV2 [21], DeiT3 [22], and PVTv2 [23], which are SOTA transformer-based architec- tures employed to analyze how effectively global attention mechanisms capture spatial relationships of subcellular structures. 4) Efficient Vision Transformers (§): Mobile- ViTv2 [24], LeViT [25], and EfficientFormerV2 [26]—lightweight transformer variants optimized for computational efficiency—are included to assess the performance of SAPLoc under resource-constrained conditions.

This multi-category comparison framework not only verifies the superiority of SAPLoc over task-specific models but also demonstrates its competitiveness against general computer vision models across different architectural paradigms, thereby validating the universality and robustness of the proposed method in handling high-resolution protein localization imagery.

2.4 Comparison on Vislocas Dataset

Table 1 Example-level metric performance comparison of SAPLoc against existing models across different input resolutions. Bolded values indicate the best performance for each input size, with red bolded values denoting the top results across all resolutions.

Model Name	Input Resolutions	EA(%)↑	EP(%)↑	ER(%)↑	EF1(%)↑	SA(%)↑
ResNet18 [18]†	224^2	62.59	71.99	62.59	66.96	53.99
ResNet50 [18]†		61.49	70.80	62.00	66.11	52.46
DenseNet121 [19]†		63.18	71.08	66.36	**68.64**	53.14
ViT-B [20]‡		61.40	70.80	61.40	65.77	52.80
SwinV2-B [21]‡		63.13	72.16	63.19	67.37	54.84
DeiT3-B [22]‡		61.74	70.80	62.03	66.12	52.97
PVTv2-B5 [23]‡		61.40	70.80	61.40	65.77	52.80
MobileViTv2-2.0 [24]§		56.30	61.27	69.86	65.29	40.58
LeViT [25]§		59.59	66.85	68.11	67.47	47.37
EfficientFormerV2-S0 [26]§		50.55	57.00	60.13	58.53	38.71
EfficientFormerV2-L [26]§		54.93	59.85	**70.09**	64.57	39.22
SAPLoc (Ours)		**63.79**	**72.41**	64.26	68.09	**55.35**
ResNet18 [18]†	512^2	64.18	73.26	65.11	68.95	54.84
ResNet50 [18]†		64.69	73.43	65.56	69.27	55.86
DenseNet121 [19]†		63.47	72.81	64.18	68.22	54.16
ViT-B [20]‡		63.78	71.99	63.78	67.64	56.37
SwinV2-B [21]‡		63.61	72.84	64.12	68.20	54.67
DeiT3-B [22]‡		62.20	70.88	62.62	66.50	53.65
PVTv2-B5 [23]‡		61.40	70.80	61.40	65.77	52.80
MobileViTv2-2.0 [24]§		63.10	71.19	64.20	67.52	54.50
LeViT [25]§		61.04	68.17	65.28	66.69	50.59
EfficientFormerV2-S0 [26]§		60.06	67.53	62.51	64.92	50.93
EfficientFormerV2-L [26]§		59.08	66.24	62.39	64.26	49.41
DeePSLoc [16]£		61.91	71.31	61.91	66.28	53.31
SAPLoc (Ours)		**65.37**	**73.54**	**68.34**	**70.84**	**57.56**
ResNet18 [18]†	1500^2	64.71	74.19	64.88	69.23	55.86
ResNet50 [18]†		66.33	**75.21**	68.53	71.72	56.03
DenseNet121 [19]†		64.15	73.03	65.31	68.96	55.35
SwinV2-B [21]‡		62.62	69.07	64.35	66.63	55.18
MobileViTv2-2.0 [24]§		63.71	71.65	65.20	68.27	54.84
EfficientFormerV2-S0 [26]§		63.75	72.67	64.54	68.36	54.84
EfficientFormerV2-L [26]§		64.38	72.72	65.70	69.04	55.35
SAPLoc (Ours)		**67.32**	75.16	**68.62**	**71.74**	**59.25**
ResNet18 [18]†	3000^2	64.36	71.82	65.56	68.55	56.54
DenseNet121 [19]†		65.58	74.73	67.60	70.99	55.35
GraphLoc [17]£		56.17	56.48	63.89	59.96	48.15
MobileViTv2-2.0 [24]§		66.16	73.26	67.03	70.01	58.91
Vislocas [3]£		67.12	**76.46**	68.59	72.31	57.05
SAPLoc (Ours)		**69.09**	76.09	**70.57**	**73.23**	**61.20**

To ensure fair and comprehensive evaluation, we re-trained and re-evaluated all computer vision models across four Image resolutions, while retaining the original image resolutions specified in the publications of PSL-specific models. For the Vislocas model, whose original paper notes it can be trained from scratch on full-size IHC images by cropping image edges (rather than downsampling), we compare it with SAPLoc on full-size inputs, maintaining consistency with its training setup.

2.4.1 Example-Level Performances

As shown in Table 1, across all example-level metrics, SAPLoc outperforms all base- lines, with advantages amplified at higher resolutions, aligning with the observed resolution trend. At 3000 × 3000 pixels (full resolution), a SA of 61.20% is achieved by SAPLoc, which is 4.15% higher than Vislocas (57.05%), the strongest PSL-specific baseline. This gap is widened as resolution increases: the SA of SAPLoc is improved by 5.85% (from 55.35% at 224 × 224 to 61.20% at 3000 × 3000), outpacing the gains achieved by competing models.

In other example-level metrics, the competitive results are established by SAPLoc at full resolution: the highest EA of 69.09%, which is 1.97% higher than Vislocas; the highest ER of 70.57%, which is 1.98% higher than Vislocas; and the highest EF1 of 73.23%, which is 0.92% higher than Vislocas. Even at intermediate resolutions (512 × 512 and 1500 × 1500), the leading performance of SAPLoc is maintained, with EF1 gains of 1.57% and 0.02% over the closest baselines (ResNet50 and ResNet50, respectively), highlighting robustness across scales. While the highest EP of 76.46% at full resolution is achieved by Vislocas, SAPLoc closely trails with an EP of 76.09%, demonstrating competitive performance in this metric.

2.4.2 Lable-Level Performances

As demonstrated in Fig. 2, at the label level evaluation, dominance is continued by SAPLoc, achieving the highest LA of 85.37%. The LF1 of 68.29% is 0.35% higher than Vislocas (67.94%), confirming strong performance in fine-grained label prediction. Notably, while the Label Recall (LR) of SAPLoc is slightly lower than GraphLoc's, the Label Precision (LP) of 75.2% far exceeds GraphLoc's 43.75%, indicating a superior balance between precision and recall that is critical for distinguishing morphologically similar organelles.

2.5 Ablation Study

2.5.1 Impact of Sparse Self-Attention

To isolate the impact of SSA, we conduct a controlled ablation study on the Vislocas dataset (Fig. 3). By setting the sparsity ratio of the SSA module to 1 (thereby reduc- ing SSA to standard global attention), the computational complexity of our model increases dramatically. Quantitatively, replacing SSA with global attention elevates the model's FLOPs from 31.59G to 40.76G. Conversely, reintroducing SSA enables efficient processing of higher-resolution inputs, achieving a SA of 57.56% at 512 × 512—1.36%

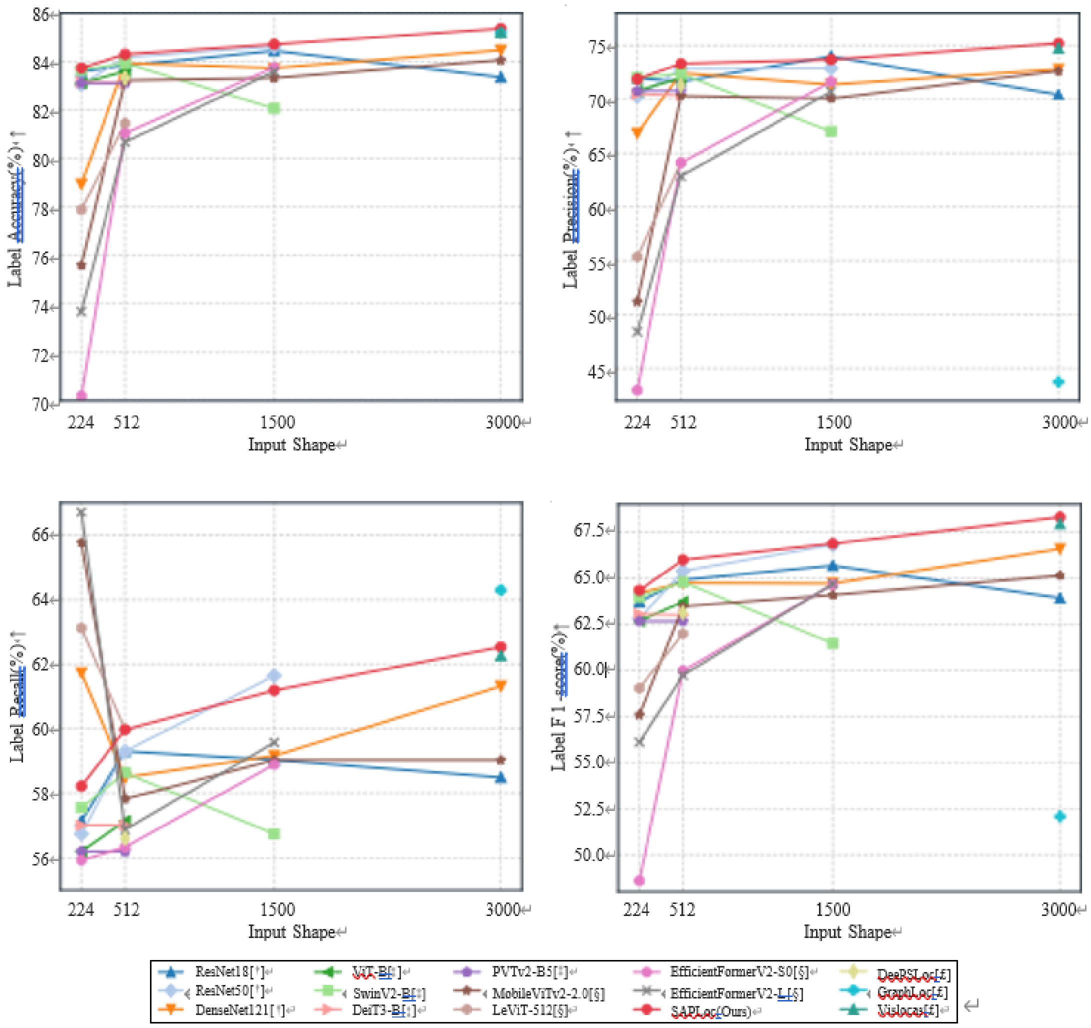


Fig. 2 Label-level metric performance comparison between SAPLoc and existing models across different input resolutions. Zoom in for better visualization.

higher than the global attention baseline. Crucially, this improvement occurs alongside reduced computational overhead, underscoring the dual role of SSA in enhancing both efficiency and accuracy. Other metrics (e.g., EF1, EP) remain stable or improve slightly, further validating the efficacy of SSA in maintaining localization performance while optimizing resource utilization.

2.5.2 Analysis of Sparse Ratios

To evaluate the sensitivity of the SAPLoc model to sparsity parameters, we conducted a comparative experiment with 512^2 input size to evaluate its performance under different key configurations (Table 2). Under the baseline configuration stage 4 sparsity rate[S_5=2, S_6=1], the model achieved optimal performance across core metrics: SA of

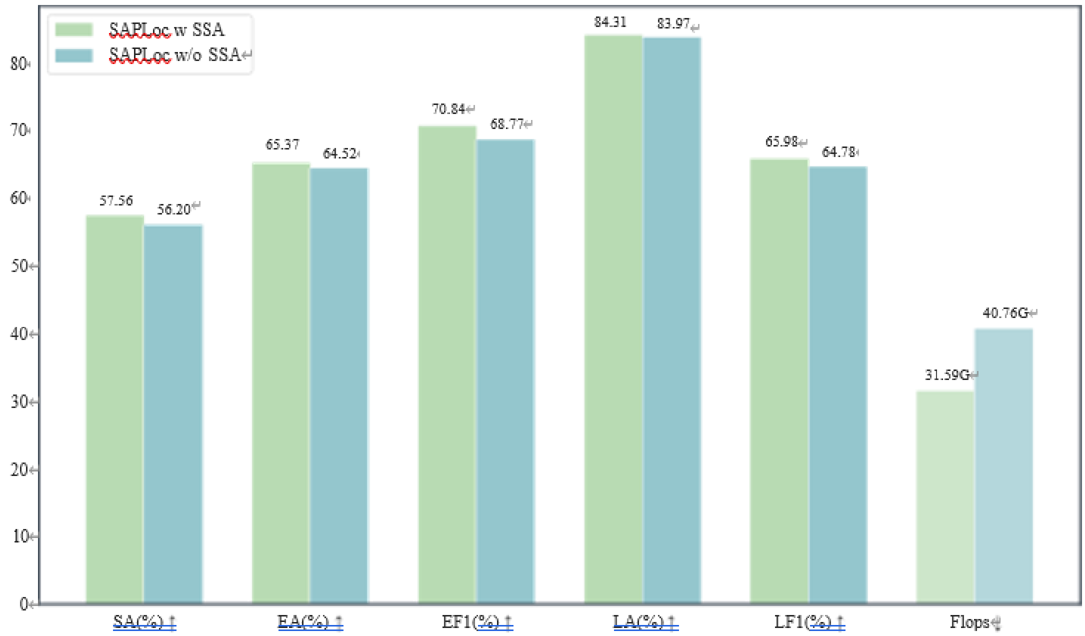


Fig. 3 Performance comparison of models with and without SSA. To assess the efficacy of SSA, comprehensive experiments are conducted at 512^2 resolution.

57.56%, EF1 of 70.84%, and LF1 of 65.98%. This indicates that a balanced parameter configuration enables effective identification of key information within complex data.

To evaluate how SAPLoc responds to sparse ratios, comparative experiments with an input size of 512^2 were conducted to assess the model's performance under differ- ent configurations, as shown in Table 2. Under the baseline configuration with stage 4 sparse ratios [S_5 = 2, S_6 = 1], the model achieved optimal performance across core metrics: SA of 57.56%, EF1 of 70.84%, and LF1 of 65.98%. This indicates that a bal- anced configuration of sparse ratios enables effective identification of key information within complex data.

Table 2 Comparison of different sparse ratios. Bolded values indicate the best performance per metric.

Sparsity rate	SA(%)↑	EA(%)↑	EF1(%)↑	LA(%)↑	LF1(%)↑
S_5=4, S_6=2	55.86	64.37	68.72	83.84	64.74
S_5=4, S_6=1	56.37	64.60	68.91	83.97	64.83
S_5=1, S_6=1	57.05	**66.47**	69.71	84.01	64.99
S_5=2, S_6=1	**57.56**	65.37	**70.84**	**84.31**	**65.98**

Compared to the final version of SAPLoc, reducing the stage 4 sparse ratios to [1, 1] resulted in performance degradation across metrics: SA decreased by 0.51% (57.05% vs. 57.56%), EF1 decreased by 1.13% (69.71% vs. 70.84%), and LF1 decreased by

0.99% (64.99% vs. 65.98%). Increasing the sparse ratios to [4, 1] led to more significant performance declines: SA decreased by 1.19% (56.37% vs. 57.56%), EF1 decreased by 1.93% (68.91% vs. 70.84%), and LF1 decreased by 1.15% (64.83% vs. 65.98%). These results confirm the importance of moderate sparse ratios: excessively low ratios retain excessive irrelevant details, introducing noise that impedes model learning, while excessively high ratios cause loss of critical information and disrupt feature coherence, particularly impairing label prediction accuracy.

When a fully sparse attention mechanism was used with sparse ratios [4, 2], perfor- mance degradation was most pronounced: SA decreased by 1.7% (55.86% vs. 57.56%), EF1 decreased by 2.21% (68.72% vs. 70.84%), and LF1 decreased by 1.24% (64.74% vs. 65.98%). This underscores the indispensable role of global attention integrated with low-sparse-ratio SSA, as the fully sparse mechanism fails to effectively establish long-range dependencies, severely impacting label prediction capability.

3 Discussion

As quantitatively shown in Table 3, superior advantages of SAPLoc are demonstrated in balancing model performance and computational complexity, along with efficient scalability across high resolutions.

3.1 Balance Between Accuracy and Computational Efficiency

At low resolution (224^2), a SA of 55.35% is achieved by SAPLoc with 5.31 GFLOPs. This outperforms SwinV2-B, which achieves 54.84% SA at the same resolution but requires 20.37 GFLOPs—meaning operation at 26.1% of SwinV2-B's computational cost is realized by SAPLoc while delivering higher accuracy.

This efficiency advantage is amplified at high resolutions. At 3000^2, the high- est SA of 61.20% is achieved by SAPLoc with 727.79 GFLOPs. In contrast, 58.91% SA is reached by MobileViTv2-2.0 at the same resolution, which requires 994.51 GFLOPs—36.7% higher computational cost than SAPLoc for lower accuracy. Even Vislocas, which operates at extremely low cost (12.24 GFLOPs), lags significantly in accuracy (57.05% SA), highlighting the unique ability of SAPLoc to balance precision and efficiency under computational constraints.

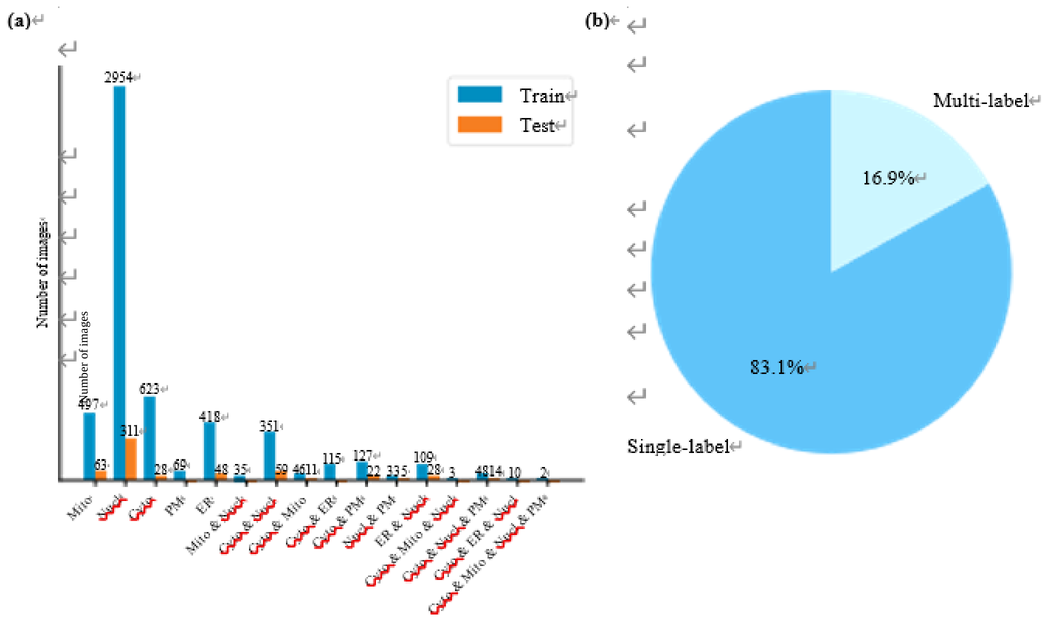
3.2 Controlled Computational Growth

Across the resolution range from 224^2 to 3000^2, an approximate 137-fold increase in FLOPs is observed for SAPLoc (from 5.31G to 727.79G). This growth is significantly lower than the theoretical quadratic scaling (179 times, proportional to pixel area growth) and far below the 176.7-fold increase of MobileViTv2-2.0 (from 5.63G to 994.51G). Even at intermediate scales, this efficiency holds. From 224^2 to 512^2, a 5.95-fold growth in FLOPs is exhibited by SAPLoc (from 5.31G to 31.59G), comparable to the 6.10-fold growth of ViT-B (from 17.73G to 108.22G) but from a much lower baseline. This results in 31.59G FLOPs for SAPLoc versus 108.22G for ViT-B at 512^2, with higher accuracy achieved by SAPLoc (57.56% vs. 56.37% SA).

Table 3 Computational efficiency and model complexity comparison on competing models. Metrics (↑ indicates higher better) and complexity metrics (↓ indicates lower better).

Model Name	Input Resolutions	SA(%)↑	Flops↓	MACs↓	Params↓
ResNet18 [18]t		53.99	1.82G	0.91G	11.18M
ResNet50 [18]t		52.46	4.11G	2.06G	23.53M
DenseNet121 [19]t		53.14	2.86G	1.43G	6.96M
VIT-B [20]‡		52.80	17.73G	8.86G	92.89M
SwinV2-B [21]‡		54.84	20.37G	10.19G	86.9M
DeiT3-B [22]‡	224^2	52.97	17.58G	8.79G	85.82M
PVTv2-B5 [23]‡		52.80	11.76G	5.88G	81.45M
MobileViTv2-2.0 [24]§		40.58	5.63G	2.81G	17.43M
LeViT [25]§		47.37	5.64G	2.82G	94.16M
EfficientFormerv2-S0 [26]§		38.71	0.41G	0.20G	3.25M
EfficientFormerv2-L [26]§		39.22	2.59G	1.30G	25.56M
SAPLoc(Ours)		55.35	5.31G	2.65G	33.33M
ResNet18 [18]t		54.84	9.50G	4.75G	11.18M
ResNet50 [18]t		55.86	21.47G	10.74G	23.53M
DenseNet121 [19]t		54.16	14.97G	7.48G	6.96M
ViT-B [20]‡		56.37	108.22G	54.11G	93.53M
SwinV2-B [21]‡		54.67	81.33G	40.67G	86.9M
DeiT3-B [22]‡		53.65	107.13G	53.56G	86.46M
PVTv2-B5 [23]‡	512^2	52.80	69.64G	34.82G	81.45M
MobileViTv2-2.0 [24]§		54.50	28.87G	14.44G	17.43M
LeViT [25]§		50.59	34.78G	17.39G	94.21M
EfficientFormerv2-S0 [26]§		50.93	2.56G	1.28G	3.26M
EfficientFormerv2-L [26]§		49.41	14.57G	7.28G	25.59M
DeePSLoc [16]£		53.31	12.76G	6.38G	28.14M
SAPLoc(Ours)		57.56	31.59G	15.80G	33.33M
ResNet18 [18]t		55.86	81.84G	40.92G	11.18M
ResNet50 [18]t		56.03	185.05G	95.52G	23.53M
DenseNet121 [19]t		55.35	127.61G	63.80G	6.96M
SwinV2-B [21]‡		55.18	731.61G	365.81G	86.9M
MobileViTv2-2.0 [24]§	1500^2	54.84	249.24G	124.62G	17.43M
EfficientFormerV2-S0 [26]§		54.84	57.40G	28.70G	3.39M
EfficientFormerV2-L [26]§		55.35	209.57G	104.79G	25.84M
SAPLoc (Ours)		59.25	100.35G	50.17G	33.37M
ResNet18 [18]t		56.54	326.98G	163.49G	11.18M
DenseNet121 [19]t		55.35	512.87G	256.44G	6.96M
MobileViTv2-2.0 [24]§	3000^2	58.91	994.51G	497.25G	17.43M
Vislocas [3]£		57.05	12.24G	6.12G	0.87M
SAPLoc (Ours)		61.20	727.79G	363.90G	33.37M

Together, these results confirm that a superior balance between accuracy and computational efficiency is enabled by the sparse design of SAPLoc, with efficient scalability across resolutions, critical for practical deployment in high-resolution PSL tasks where computational resources are limited.



g. 4 Distribution of Labels in Vislocas Dataset. (a) Distribution of Label Types in the Vislocas Dataset, which includes a total of 5 label classes. Specifically, the training set comprises 16 label combinations with 5440 images, while the test set contains 10 combinations with 589 images. (b) The distribution ratio of single labels to multiple labels in the Vislocas dataset.

4 Methods

4.1 Datasets

In this study, the performance of SAPLoc is evaluated using the Vislocas dataset. Derived from the HPA database through screening, the Vislocas dataset contains 6029 high-quality stained images of normal tissues, covering 548 different proteins from the latest version (Version 24) of the HPA database. A new annotation protocol is adopted by the Vislocas dataset: under the premise of identical protein, antibody, and source, subcellular localization annotation information from high-resolution flu- orescence images is used to replace staining labels. These images involve five basic organelle categories: cytoplasm (Cyto), endoplasmic reticulum (ER), mitochondria (Mito), nucleus (Nucl), and plasma membrane (PM). Fig. 4(a) intuitively presents the label distribution of the training and test sets in the Vislocas dataset, enabling clear observation of sample count differences among the five organelle categories in both sets. Additionally, the Vislocas dataset includes both single-label and multi-label instances. As shown in Fig. 4(b), the proportion of single-label instances is 83.11%, and that of multi-label instances is 16.89%. The dataset exhibits a significant charac- teristic of class imbalance: there are 3962 protein samples with nuclear localization, while only 320 protein samples with plasma membrane localization, resulting in a quantity gap of over 12 times between the two. The number of samples in other cate- gories (cytoplasm, endoplasmic reticulum, and mitochondria) also varies, collectively contributing to this unbalanced distribution.

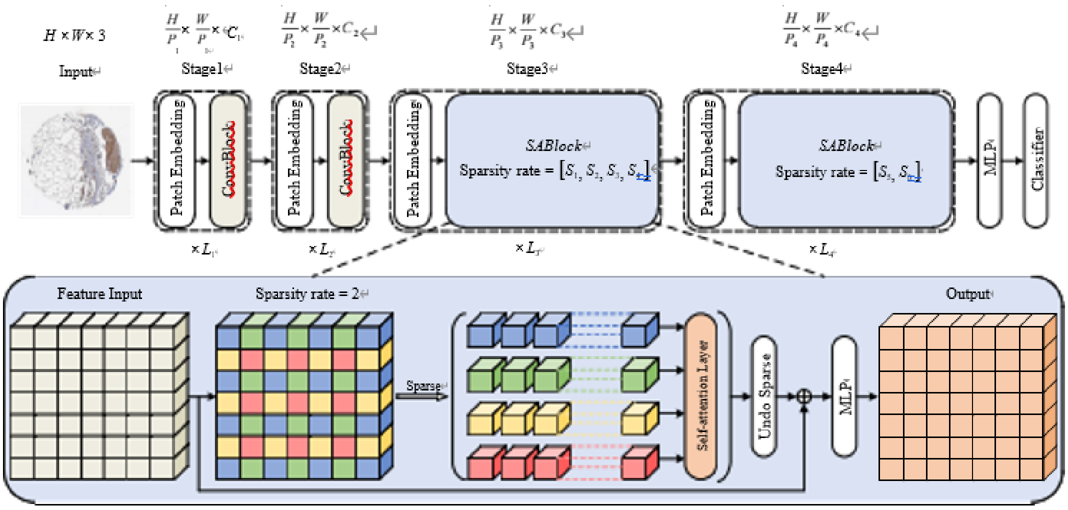


Fig. 5 Architecture of the SAPLoc framework. The pipeline comprises four stages, with sparse attention modules in Stage 3 and Stage 4 enabling localized feature interaction while reducing com- putational overhead.

4.2 Overview of SAPLoc

The proposed SAPLoc framework leverages a synergistic architecture integrating sparse attention mechanisms with hierarchical feature encoding, specifically designed to process IHC images through four sequential stages. As illustrated in Fig. 5, for an input image $\mathbf{I} \in \mathrm{R}^{H\times W\times 3}$, the framework first performs initial feature extraction and dimensionality transformation via two successive patch embedding and Convolutional Blocks (ConvBlocks) operations in Stage 1 and Stage 2. Each ConvBlock consists of a convolutional layer followed by a normalization and activation function, collec- tively forming a foundational visual representation of the input. Specifically, the stage 1 processes $\mathbf{I}$ to produce feature map $\mathbf{X_1} \in \mathrm{R}^{\frac{H}{P_1}\times\frac{W}{P_1}\times C_1}$, and the stage 2 generates $\mathbf{X_2} \in \mathrm{R}^{\frac{H}{P_2}\times\frac{W}{P_2}\times C_2}$. Subsequently, the extracted feature maps are fed into two sparse attention modules in Stage 3 and Stage 4, each configured with distinct sparsity rates (S) to adaptively emphasize critical local features while suppressing irrelevant global semantic informa- tion: the stage 3 outputs $\mathbf{X_3} \in \mathrm{R}^{\frac{H}{P_3}\times\frac{W}{P_3}\times C_3}$ through successive sparsity rate reductions and self-attention interactions, initiating global context integration; the stage 4 fur- ther integrates global correlations and strengthens cross-region feature dependencies to generate $\mathbf{X_4} \in \mathrm{R}^{\frac{H}{P_4}\times\frac{W}{P_4}\times C_4}$. This selective focus enhances the model's capability to capture subtle protein localization patterns within cellular structures. After pro- cessing through the sparse attention stages, the refined features are passed through a multi-layer perceptron (MLP) to generate the final pixel-level prediction mask.

4.2.1 Sparse self-attention

Traditional deep learning models for visual tasks often prioritize global semantic features through dense self-attention, where every spatial token interacts with all others. However, this global interaction paradigm is suboptimal for IHC image analysis, as it introduces redundant computations and dilutes focus on critical local patterns (e.g., protein aggregates within cells).

To address this, SAPLoc employs SSA, introducing a key hyperparameter termed the "sparsity rate" (S). For an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, instead of computing attention across the entire spatial domain, the feature map is partitioned into $S \times S$ non-overlapping sub-blocks, each of size $\frac{H}{S} \times \frac{W}{S} \times C$. Attention is computed exclusively within each sub-block, as visualized in Fig. 5, where interactions are restricted to tokens within the same color-coded region.

For a given sub-block $\mathbf{Z}_S \in \mathbb{R}^{\frac{H}{S} \times \frac{W}{S} \times D}$, query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices are generated via learnable projections shared across all sub-blocks:

$$\mathbf{Q}_S = \mathbf{Z}_S \mathbf{W}_Q, \quad \mathbf{K}_S = \mathbf{Z}_S \mathbf{W}_K, \quad \mathbf{V}_S = \mathbf{Z}_S \mathbf{W}_V \tag{1}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times D}$ denote projection weights. The intra-block attention is then calculated as:

$$\text{Attention}_S = \text{Softmax}\left(\frac{\mathbf{Q}_S \mathbf{K}_S^\top}{\sqrt{d_k}}\right) \mathbf{V}_S \tag{2}$$

Here, $\sqrt{d_k}$ stabilizes gradient propagation, and softmax is applied row-wise to the $N_S \times N_S$ similarity matrix $(N_S = \frac{HW}{S^2})$.

This design strengthens local feature interactions, mitigates distraction from irrelevant global semantics, and eliminates redundant cross-block computations. Com- pared to global attention with $O((HW)^2)$ complexity, sparse attention reduces this to $O\left(\left(\frac{HW}{S^2}\right)^2\right)$, enabling efficient processing of high-resolution IHC images while preserving critical local patterns.

4.2.2 Zero-bounded Log-sum-exp & Pairwise Rank-based Loss

To address the multi-label classification task of PSL, we adopted the zero-bounded log-sum-exp & pairwise rank-based (ZLPR) [27] loss. This loss function combines the concepts of log-sum-exp and pairwise ranking by computing logarithmic func- tions involving exponential terms for the set of positive labels and corresponding logarithmic functions for the set of negative labels, thereby constraining the output scores of positive and negative classes, respectively, to optimize label prediction. This design addresses the uncertainty in the number of target labels while accounting for correlations between labels. The ZLPR loss function is as follows:

$$\mathcal{L}_{\text{ZLPR}} = \log\left(1 + \sum_{i \in \Omega_{\text{neg}}} e^{s_i}\right) + \log\left(1 + \sum_{j \in \Omega_{\text{pos}}} e^{-s_j}\right) \tag{3}$$

where $\mathbf{s} \in R^n$ denotes the model's class predictions, and Ω_{neg} , Ω_{pos} reprensent the index sets of positive and negative class, respectively. A key characteristic of ZLPR Loss is its ”zero-boundary” property: during prediction, all positive class scores $s_i > 0$ and all negative class scores $s_j < 0$. Through the zero threshold ($s_0 = 0$), it achieves adaptive targeting of label quantities, eliminating the need for manual thresh- old setting or additional modules to accommodate the uncertainty in label counts. Furthermore, ZLPR Loss captures inter-label correlations by leveraging the joint distri- bution of labels, which is particularly crucial for subcellular structures with dependent relationships.

5 Conclusion

In this study, we present SAPLoc, a novel framework designed to address the criti- cal challenges of computational efficiency and fine-grained feature preservation in PSL prediction from high-resolution IHC images. By integrating SSA with hierarchical multi-scale encoding, SAPLoc bridges the gap between the demand for high-resolution biological detail and practical computational constraints, offering a scalable solution for PSL analysis. Experimental results on the Vislocas dataset validate SAPLoc's supe- riority: At full resolution, it achieves leading performance, with significant advantages over the strongest PSL-specific baseline. This performance edge becomes more pro- nounced as resolution increases, accompanied by well-controlled computational growth that avoids the theoretical quadratic scaling expected from expanded pixel areas. Notably, SAPLoc maintains strong accuracy across all tested resolutions, underscor- ing its unique ability to effectively harness high-resolution details pivotal for precise subcellular localization. By achieving a superior balance between accuracy and compu- tational efficiency, SAPLoc advances PSL prediction from high-resolution IHC images, providing a practical tool for studying protein function and disease mechanisms. Future work will focus on enhancing multi-scale feature fusion and exploring cross-modal integration with protein sequence data to further improve localization precision in complex tissue environments.

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